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QUANTUM-IN-THE-LOOP CONTROL: MPC WITH QAOA FOR ROBOTICS

WHAT IS QAOA?

Quantum Approximate Optimization Algorithm

Goal:

- solve combinatorial optimization problems approximately

Idea:

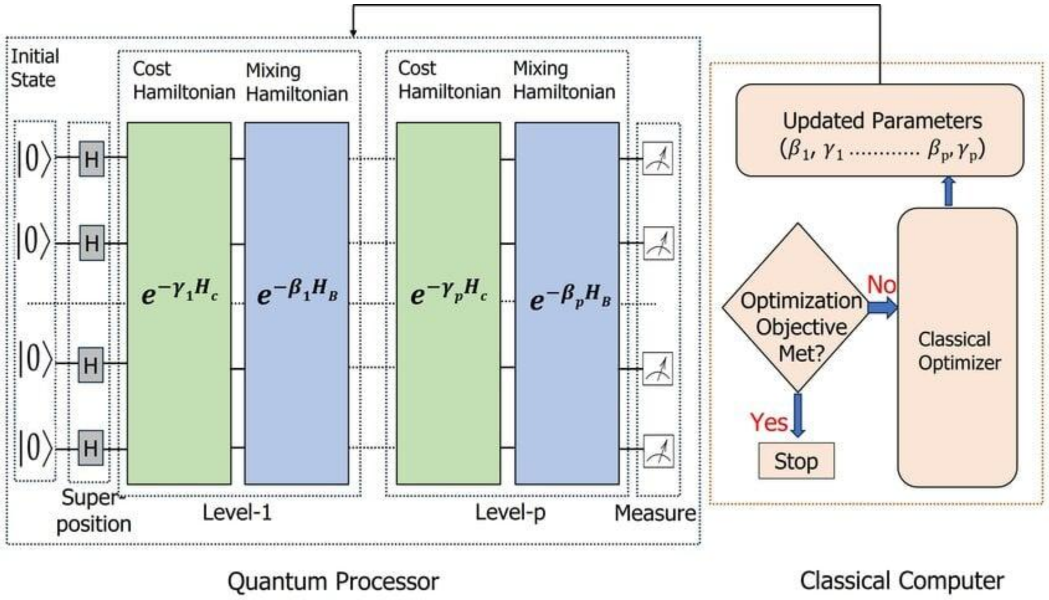
- encode the problem as a cost Hamiltonian
- alternate between:
  1. cost evolution
  2. mixing evolution
- tune circuit parameters to increase probability of low-cost bitstrings

Input form:

- QUBO / Ising optimization problems

Output:

- candidate low-cost binary solutions



$$|\psi(\gamma,\beta)\rangle = \prod_{\ell=1}^{p} e^{-i\beta_\ell H_M}\, e^{-i\gamma_\ell H_C}\, |+\rangle^{\otimes n}$$

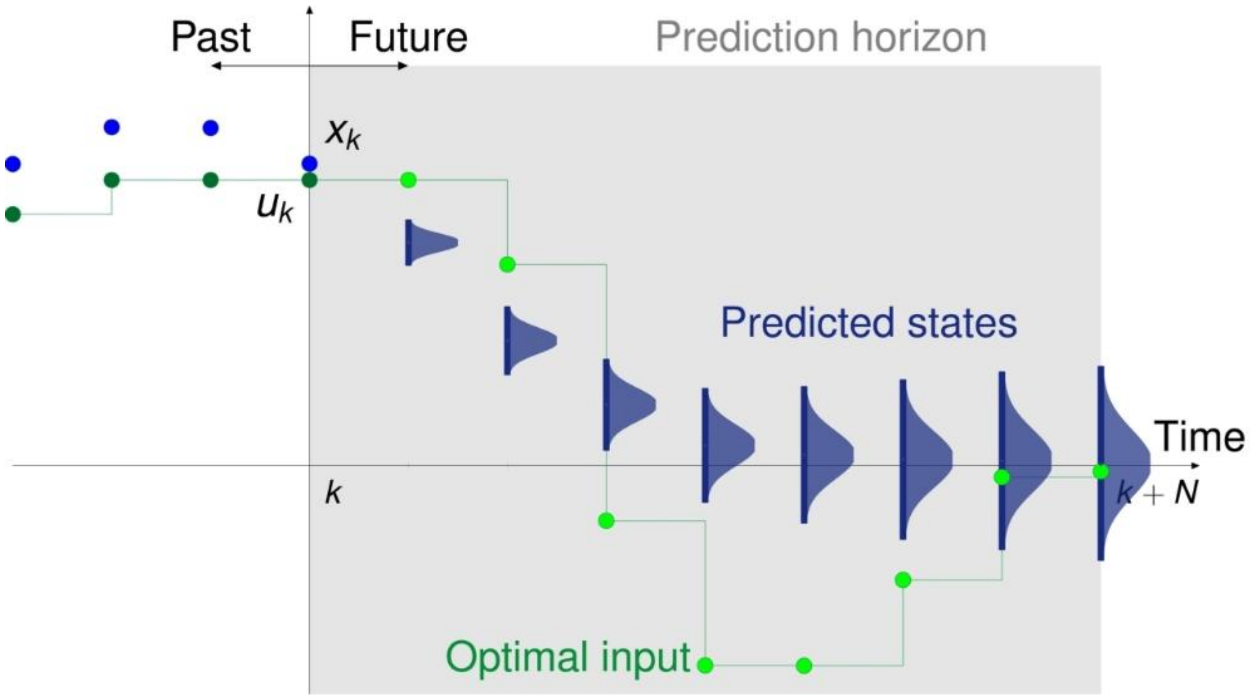
J. Ceroni, "Intro to QAOA," *PennyLane Demos*, Nov. 18, 2020. https://pennylane.ai/qml/demos/tutorial_qaoa_intro

WHAT IS MPC?

At each step:

1. measure current state
2. predict future states using a model
3. optimize a short sequence of future controls
4. apply only the first control
5. repeat

Key idea:
plan ahead, execute one step, replan



$$x_k+1=Ax_k+Bu_k$$

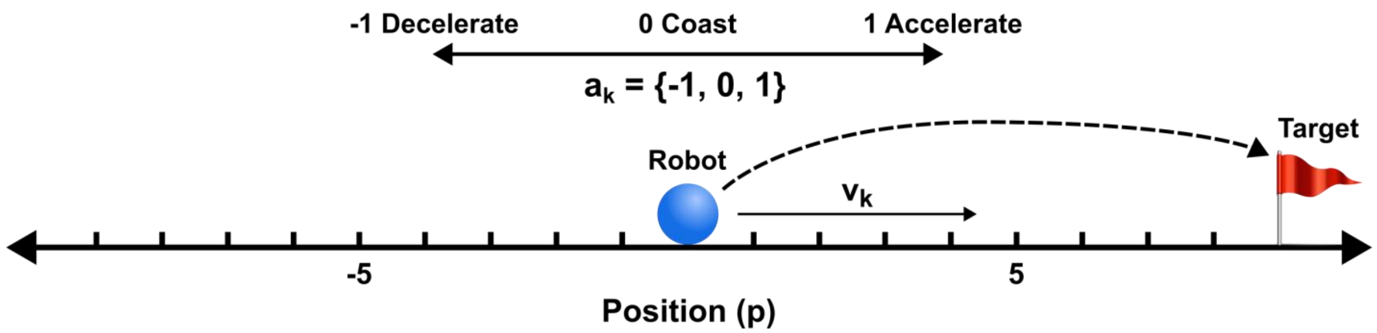
MathWorks, “What is Model Predictive Control? - MATLAB & Simulink,” *www.mathworks.com*.
https://www.mathworks.com/help/mpc/gs/what-is-mpc.html

ROBOTICS EXAMPLE

Goal:

1. Reach target
2. Minimize effort
3. Obey constraints

An MPC problem could also include state constraints like velocity bounds or obstacle avoidance.



1D Point Robot:

$$x_k = \begin{bmatrix} p_k \\ v_k \end{bmatrix}$$

Dynamics:

$$p_{k+1} = p_k + \Delta t v_k$$
$$v_{k+1} = v_k + \Delta t a_k$$

Discrete Control:

$$a_k \in \{-1, 0, 1\}$$

FROM MPC TO QAOA

MPC Problem → Discretized (Combinatorial) Form → QUBO Problem → QAOA Solver with Ising Hamiltonian

Binary Control Encoding

$$z_{k,-1}, z_{k,0}, z_{k,+1} \in \{0,1\}$$

$$z_{k,-1} + z_{k,0} + z_{k,+1} = 1$$

$$a_k = -z_{k,-1} + z_{k,+1}$$

What this means:

- one binary variable per action
- exactly one action chosen
- binary variables recover the control input

Add penalties for one-hot constraints

$$J(z) = J_{MPC}(z) + \lambda \sum_k (z_{k,-1} + z_{k,0} + z_{k,+1} - 1)^2$$

Quadratic unconstrained binary optimization (QUBO)

$$\min_{z \in \{0,1\}^m} z^T Hz + g^T z + c$$

Binary-to-spin change of variables

$$z_i = \frac{1 - s_i}{2}, \qquad s_i \in \{-1, +1\}$$

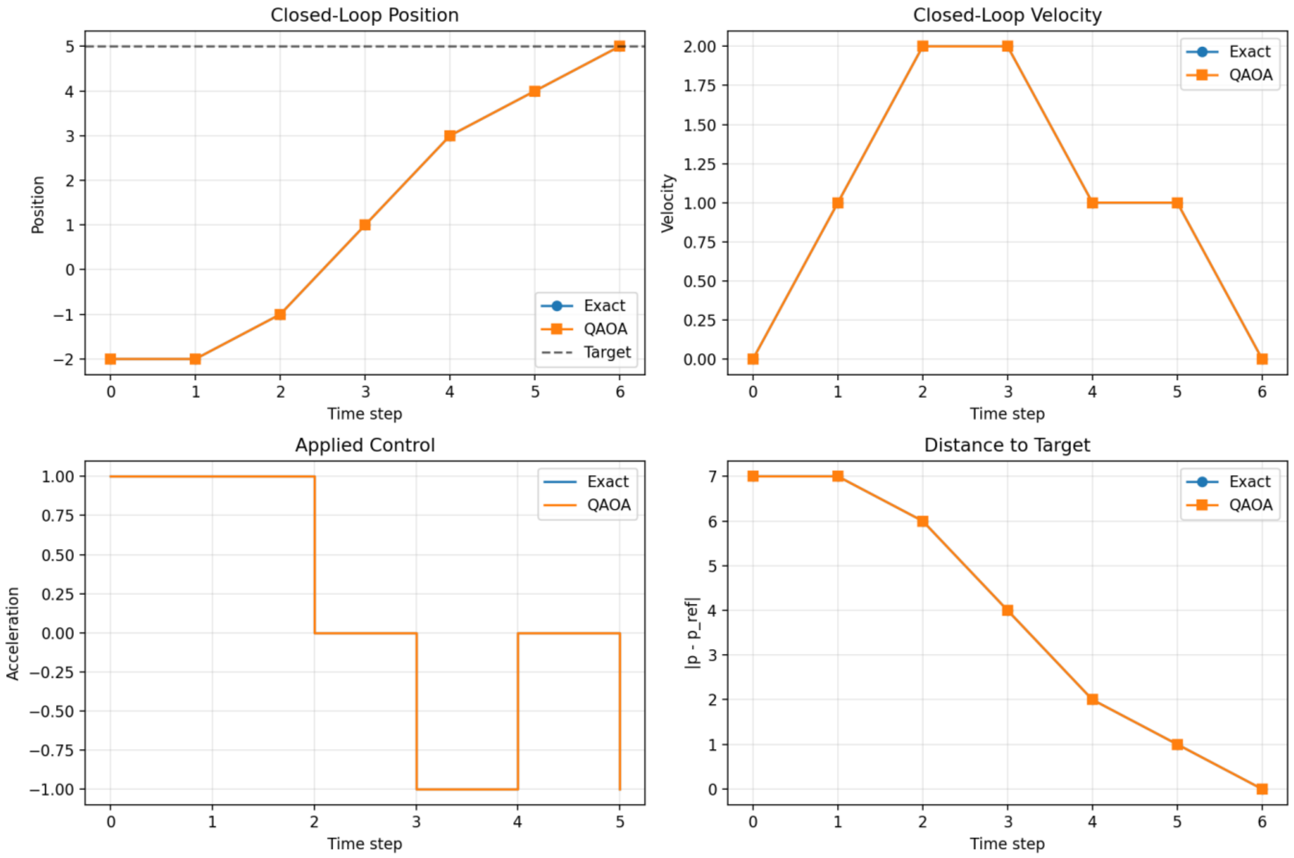
Ising Energy / Cost Hamiltonian

$$E(s) = \sum_i h_i s_i + \sum_{i<j} J_{ij} s_i s_j + const$$

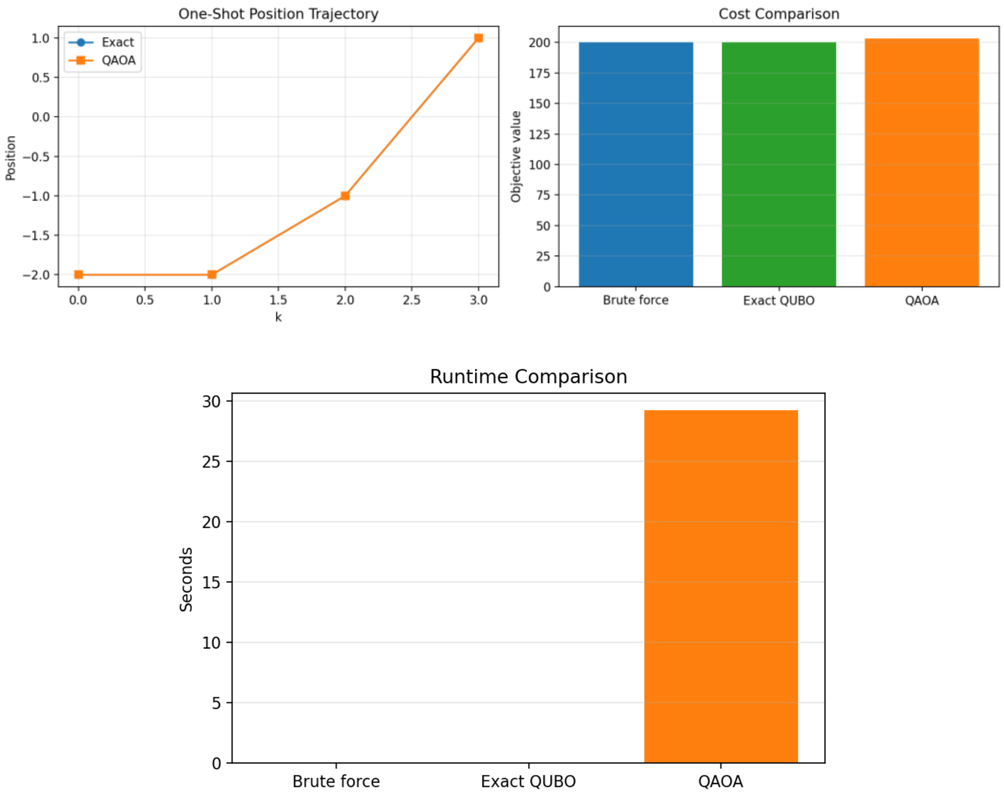
$$\hat{H}_c = \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j$$

D. Inoue and H. Yoshida, "Model Predictive Control for Finite Input Systems using the D-Wave Quantum Annealer," Scientific Reports, vol. 10, no. 1, Jan. 2020, doi: https://doi.org/10.1038/s41598-020-58081-9.

PERFORMANCE COMPARISON USING QISKIT



Closed-Loop Results



One-Shot Results

LIMITATIONS AND FUTURE WORK

Still difficult today, but several hardware and systems research directions could make quantum-in-the-loop robotic control more plausible over time.

Fault-Tolerant QC Hardware

- Lower error rates
- More reliable logical qubits
- Deeper circuits become possible
- Larger optimization problems become more realistic

IBM Quantum Roadmap/Starling
200 Logical Qubits and 100m gates by 2029

Novel Actuators

- Less interference near quantum hardware
- Better fit for sensitive environments
- Could enable robots to operate near QCs
- Useful for specialized low-noise systems

Piezoelectric actuators for Robotics
No EM noise - can enable operation in the same environment as a QC

Cryogenic + Low-Latency Integration

- Faster quantum-classical feedback
- Less overhead from support electronics
- Better integration inside cryogenic systems
- More realistic hybrid control loops

IBM Cryo-CMOS Control System
Hybrid cryopgenic/room-temperature architecture